



APPROXIMATION ALGORITHM

PROF. PALASH DEY

Department of Computer Science and Engineering
IIT Kharagpur

PRE-REQUISITES : Knowledge of algorithm design principles.

INTENDED AUDIENCE : Under-graduate and post-graduate students and faculty members.

INDUSTRY SUPPORT : All software companies especially Google, Microsoft, etc.

COURSE OUTLINE :

Many real-world problems are NP-complete. Hence, they are unlikely to admit a polynomial-time algorithm. In this course, we will study various techniques to design efficient algorithms to compute an approximately optimal solutions.

ABOUT INSTRUCTOR :

Prof. Palash Dey is an Assistant Professor in the Department of Computer Science and Engineering at Indian Institute of Technology, Kharagpur since 2018. Before joining IIT Kharagpur, he was a post-doctoral INSPIRE faculty in TIFR, Mumbai for one year. He finished his Ph.D. and M.E. from the Department of Computer Science and Automation at Indian Institute of Science, Bangalore in 2017 and 2013 respectively. Prior to that, he finished his B.E. from the Department of Computer Science and Engineering at Jadavpur University in 2010. His primary field of research is algorithmic game theory. He is broadly interested in theoretical computer science.

COURSE PLAN :

Week 1: Review of NP-Completeness

Week 2: Approximation algorithm for the vertex cover, set cover, and traveling salesman problem

Week 3: Approximation algorithm for the Knapsack problem and the job scheduling problems

Week 4: Basics of linear programming

Week 5: Deterministic rounding: set cover, vertex cover problems

Week 6: Deterministic rounding: steiner tree, facility location problems

Week 7: Randomized rounding: max-SAT problem

Week 8: Randomized rounding: steiner tree, facility location problems

Week 9: Primal-dual method: set cover

Week 10: Primal-dual method: steiner tree

Week 11: Semidefinite programming: max cut

Week 12: Hardness of approximation: travelling salesman problem, job scheduling